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20 / 20 submitted

Q1. Union of Sets

Score: 5.00 TC: 1/1 passed

CODE

```
a={1,2,3}
b={3,4,5}
print(str(a|b).replace(" ", ""))
```

OUTPUT

{1,2,3,4,5}

Q2. Intersection

Score: 5.00 TC: 1/1 passed

CODE

```
a={1,2,3}
b={2,3,4}
print(str(a&b).replace(" ", ""))
```

OUTPUT

{2,3}

Q3. Difference

Score: 5.00 TC: 1/1 passed

CODE

```
a={1,2,3}
b={2,3,4}
print(str(a-b).replace(" ", ""))
```

OUTPUT

{1}

Q4. Symmetric Difference

Score: 5.00 TC: 1/1 passed

CODE

```
a={1,2,3}
b={2,3,4}
print(str(a^b).replace(" ", ""))
```

OUTPUT

{1,4}

Q5. Subset

Score: 5.00 TC: 1/1 passed

CODE

```
a={2,3}
b={1,2,3,4}
if(a.issubset(b)):
    print("Subset")
else:
    print("not a subset")
```

OUTPUT

Subset

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Q6. Superset

Score: 5.00 TC: 1/1 passed

CODE

```
a={1,2,3,4}
b={2,3}
if(a.issuperset(b)):
    print("Superset")
else:
    print("not a superset")
```

OUTPUT

Superset

Q7. Disjoint Sets

Score: 5.00 TC: 1/1 passed

CODE

```
a={1,2}
b={3,4}
if(a.isdisjoint(b)):
    print("Disjoint")
else:print("not disjoint")
```

OUTPUT

Disjoint

Q8. Remove Duplicates

Score: 5.00 TC: 1/1 passed

CODE

```
a=[1,2,2,3,3,4]
b=set(a)
print(str(b).replace(", ", ","))
```

OUTPUT

{1,2,3,4}

Q9. Common Elements

Score: 5.00 TC: 1/1 passed

CODE

```
a={1,2,3}
b={2,3,4}
c={2,5}
print(a&b&c)
```

OUTPUT

{2}

Q10. Missing Numbers

Score: 0.00 TC: 0/1 passed

CODE

```
f=set(range(1,11))
g={1,2,4,5,7,10}
m=f-g
print(str(m).replace(", ", ","))
```

OUTPUT

{8,9,3,6}

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Q11. First Repeated

Score: 5.00 TC: 1/1 passed

CODE

```
nums = list(map(int, input().split()))

seen = set()
first_repeated = None

for num in nums:
    if num in seen:
        first_repeated = num
        break
    seen.add(num)

print(first_repeated)
```

INPUT

10 20 30 20 40

OUTPUT

20

Q12. Duplicate Elements

Score: 5.00 TC: 1/1 passed

CODE

```
a=[10,20,30,20,10]
seen=set()

d=set()
for i in a:
    if i in seen:
        d.add(i)
    else:
        seen.add(i)

print(str(d).replace(" ", ","))
```

OUTPUT

{10,20}

Q13. Compare Lists

Score: 0.00 TC: 0/1 passed

CODE

```
list1 =[1,2,2,3]
list2 =[3,2,1]
print(set(list1)==set(list2))
```

INPUT

1 2 2 3
3 2 1

OUTPUT

True

Q14. Duplicate Tuples

Score: 5.00 TC: 1/1 passed

CODE

```
a= [(1,2),(1,2),(3,4)]
for i in a:
    if(a.count(i)>1):
        a.pop(a.index(i))
print(str(a).replace(" ", ","))
```

OUTPUT

[(1,2),(3,4)]

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Q15. Pair Sum

Score: 5.00 TC: 1/1 passed

CODE

```
l=[2,4,6,8,10]
t=12
s=set()
p=[]
for n in l:
    d=t-n
    if d in s:
        p.append(f"({d},{n})")
    s.add(n)
print(",".join(p))
```

OUTPUT

(4,8),(2,10)

Q16. Unique Characters

Score: 5.00 TC: 1/1 passed

CODE

```
print(len(set([i for i in input()])))
```

INPUT

Programing

OUTPUT

8

Q17. Unique Words

Score: 5.00 TC: 1/1 passed

CODE

```
a="python is easy python"
print(str(set(a.split())).replace(" ", ""))
```

OUTPUT

{'python','is','easy'}

Q18. Exactly One Set

Score: 5.00 TC: 1/1 passed

CODE

```
a={1,2,3}
b={3,4,5}
c={5,6,7}
print(str(a^b^c).replace(" ", ""))
```

OUTPUT

{1,2,4,6,7}

Q19. Menu Driven

CODE

```
set1 = set(map(int, input("Enter the first set elements: ").split()))
set2 = set(map(int, input("Enter the second set elements: ").split()))

while True:
    print("\nMenu")
    print("1. Union")
    print("2. Intersection")
    print("3. Difference (A - B)")
    print("4. Symmetric Difference")
    print("5. Exit")

    choice = int(input("\nEnter your choice: "))

    if choice == 1:
        print("\nUnion:", set1 | set2)
    elif choice == 2:
        print("\nIntersection:", set1 & set2)
    elif choice == 3:
        print("\nDifference:", set1 - set2)
    elif choice == 4:
        print("\nSymmetric Difference:", set1 ^ set2)
    elif choice == 5:
        print("\nProgram terminated.")
        break
    else:
        print("\nInvalid choice. Please try again.")
```

INPUT

```
1 2 3 4
3 4 5 6
1
2
3
4
5
```

OUTPUT

```
Enter the first set elements: Enter the second set elements:
Menu
1. Union
2. Intersection
3. Difference (A - B)
4. Symmetric Difference
5. Exit

Enter your choice:
Union: {1, 2, 3, 4, 5, 6}

Menu
1. Union
2. Intersection
3. Difference (A - B)
4. Symmetric Difference
5. Exit

Enter your choice:
Intersection: {3, 4}

Menu
1. Union
2. Intersection
3. Difference (A - B)
4. Symmetric Difference
5. Exit

Enter your choice:
Difference: {1, 2}

Menu
1. Union
2. Intersection
3. Difference (A - B)
4. Symmetric Difference
5. Exit

Enter your choice:
Symmetric Difference: {1, 2, 5, 6}

Menu
1. Union
2. Intersection
3. Difference (A - B)
```

4. Symmetric Difference
5. Exit

Enter your choice:
Program terminated.

CODE

```
students = set()

while True:
    print("Menu")
    print("1. Add Student")
    print("2. Remove Student")
    print("3. Search Student")
    print("4. Display Students")
    print("5. Count Students")
    print("6. Exit")

    choice = int(input("Enter your choice: "))

    if choice == 1:
        name = input("\nEnter student name: ")
        students.add(name)
        print("\nStudent added successfully.")

    elif choice == 2:
        name = input("\nEnter student name to remove: ")
        if name in students:
            students.discard(name)
            print("\nStudent removed successfully.")
        else:
            print("\nStudent not found.")

    elif choice == 3:
        name = input("\nEnter student name to search: ")
        if name in students:
            print("\nStudent found.")
        else:
            print("\nStudent not found.")

    elif choice == 4:
        print("\nStudents:", students)

    elif choice == 5:
        print("\nTotal number of students:", len(students))

    elif choice == 6:
        print("\nProgram terminated.")
        break

    else:
        print("\nInvalid choice.")
```

INPUT

```
1
Ravi
2
Priya
3
Ravi
2
Ravi
5
6
```

OUTPUT

```
Menu
1. Add Student
2. Remove Student
3. Search Student
4. Display Students
5. Count Students
6. Exit
Enter your choice:
Enter student name:
Student added successfully.
Menu
1. Add Student
2. Remove Student
3. Search Student
4. Display Students
5. Count Students
```

```
6. Exit
Enter your choice:
Enter student name to remove:
Student not found.
Menu
1. Add Student
2. Remove Student
3. Search Student
4. Display Students
5. Count Students
6. Exit
Enter your choice:
Enter student name to search:
Student found.
Menu
1. Add Student
2. Remove Student
3. Search Student
4. Display Students
5. Count Students
6. Exit
Enter your choice:
Enter student name to remove:
Student removed successfully.
Menu
1. Add Student
2. Remove Student
3. Search Student
4. Display Students
5. Count Students
6. Exit
Enter your choice:
Total number of students: 0
Menu
1. Add Student
2. Remove Student
3. Search Student
4. Display Students
5. Count Students
6. Exit
Enter your choice:
Program terminated.
```